Practice workbook · The stored program concept and the parts of the CPU

Von Neumann stored program idea, what instructions and memory addresses are, and the roles of the control unit, ALU and registers, including labelling them on a diagram.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-T4-03

1. Question 1 · part a.i · 1 mark

2022 JUN · Paper 01

Worked-example retry · Attempt part a.i only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

1. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 1 · part b · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part b only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2. Question 1 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 1 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

3. Question 1 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(d) Some computers use virtual memory.

Explain how virtual memory works.

(2)

(e) An optical storage device writes data onto a CD or DVD.

Describe how data is stored physically on optical media.

(2)

OLevels.com SCHOOL

(f) A disc-formatting program is an example of a type of software.

Identify the type.

(1)

- ☒ **A** Security software
- ☒ **B** Application software
- ☒ **C** Translation software
- ☒ **D** System software

(Total for Question 1 = 11 marks)

3. Question 1 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part d.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.i only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

(1)

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

(2)

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

(3)

R	S	W	NOT S	R AND W	(NOT S) OR (R AND W)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 4 · part d.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

(2)

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

(2)

1 .....

2 .....

OLevels.com SCHOOL

(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

(2)

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

(2)

Change made	Effect
Increasing the width of the data bus	
Increasing the width of the address bus	

(Total for Question 4 = 14 marks)

4. Question 4 · part d.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part d.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.ii only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

(1)

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

(2)

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

(3)

R	S	W	NOT S	R AND W	(NOT S) OR (R AND W)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 4 · part d.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

(2)

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

(2)

1 .....

2 .....

OLevels.com SCHOOL

(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

(2)

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

(2)

Change made	Effect
Increasing the width of the data bus	
Increasing the width of the address bus	

(Total for Question 4 = 14 marks)

5. Question 4 · part d.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 5 · part a · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

5 Computers can be general-purpose or specific-purpose machines.

(a) The central processing unit (CPU) is responsible for executing a program.

Figure 4 shows some of the hardware components of a computer system.

Complete the table by matching the component to the letters shown in **Figure 4**.

(5)

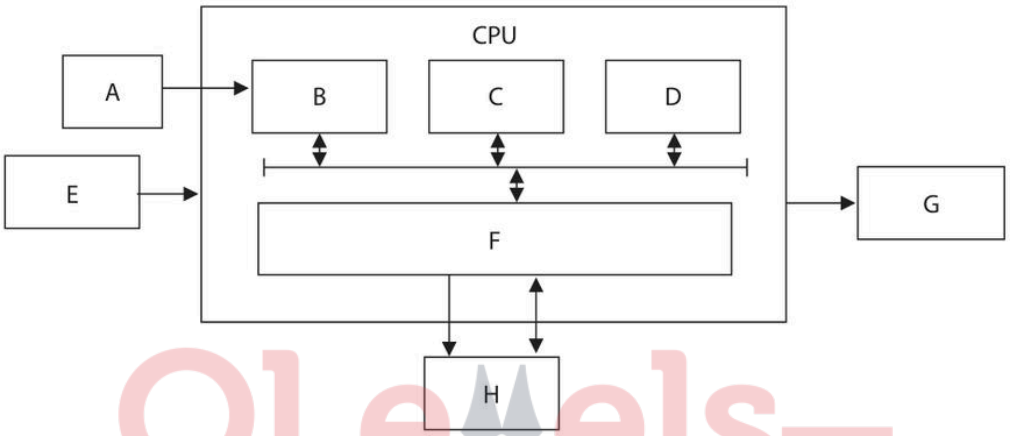


Figure 4

Component	Letter
Control unit	
Output device	
Registers	C
Main memory	
Cache	F
Input device	
Arithmetic/logic unit (ALU)	
Clock	A

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6. Question 5 · part a · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

(1)

☒ **A** Memory delete signal

☒ **B** Memory read signal

☒ **C** Memory store signal

☒ **D** Memory write signal

(iii) Clock speed is the number of pulses the CPU's clock generates per second.

State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

(1)

(c) An embedded system forms part of a larger system, device or machine.

(i) Give **two** features of an embedded system.

(2)

1 .....

2 .....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 5 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 1 · part a.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

P 7 2 5 3 8 A 0 2 2 0

7. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 1 · part b · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

2

P 7 2 5 3 8 A 0 2 2 0

8. Question 1 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL